

A scoping review of positive youth development programming for vulnerable and marginalized youth in low- and middle-income countries

Gina Chow^{a,*}, Rainier Masa, Miranda Manzanares, Neil Bilotta

Global Social Development Innovations, School of Social Work, University of North Carolina at Chapel Hill, 325 Pittsboro St. CB #3550, Chapel Hill, NC 27599-3550, USA

ARTICLE INFO

Keywords:

Positive youth development
Vulnerable and marginalized youth
Youth
Low- and Middle-Income Countries – LMICs
Programs
Scoping review
Youth programming

ABSTRACT

This scoping review aims to describe the state of Positive Youth Development (PYD) programming for vulnerable and marginalized youth in LMICs and to assess the quality of available evidence on process evaluation and program effectiveness. We systematically searched academic and grey literature to appraise existing literature on PYD as applied to research studies and programming with vulnerable and marginalized youth in low- and middle-income countries (LMICs). Our scoping review followed the four general phases of a systematic search: preparation, retrieval, screening and appraisal, and synthesis. Our search yielded 42 publications that met the inclusion criteria. We identified 38 programs across 27 countries. Most programs included in this review did not explicitly identify PYD as their underlying behavioral change approach and did not differentiate programming based on youth's levels of vulnerability and marginalization. Youth-focused programs were comparably distributed across three development sectors—health, democracy, governance, economic development, and education. Poverty, stigma, discrimination, and cultural and social norms, including gender inequities create barriers to accessing youth-focused programs. Our review also found enabling factors, such as youth-friendly policies, community involvement, congruence with cultural and social norms, and creative media use, that support vulnerable and marginalized youth's access to programs. We conclude that there remain gaps in using PYD as a conceptual framework to inform programming and as a methodological tool to standardize evaluation and impact assessment of PYD or PYD-informed programming for vulnerable and marginalized youth in LMICs.

1. Introduction

The Positive Youth Development (PYD) framework has become central to youth development programming globally. PYD, as defined by Benson and colleagues (2007), highlights the importance of building the health, cognitive, social, psychological, and emotional assets of young people through engagements and inclusion in contexts and settings that recognize these assets and provide constructive opportunities for youth to engage and build skills. Features of settings and contexts that foster such engagement include physical and psychological safety, opportunities to build skills, and supportive relationships (Lerner et al., 2009). PYD is a prosocial approach that engages youth within social, academic, community, and familial relationships by enhancing young people's strengths to facilitate positive relationships, which emphasizes leadership skills, youth competencies, abilities, and skills (Hinson et al., 2016). Researchers have found associations between PYD and positive youth outcomes (Zhou, Shek, & Zhu, 2020). The PYD framework delineates

four interrelated domains crucial to youth development—agency, assets, contribution, and an enabling environment (Hinson et al., 2016). The agency domain includes youth's perception of their ability to apply their assets to influence their decisions to make in order to receive desired outcomes (Hinson et al., 2016). Assets constitute the skills, relationships, opportunities, and values youth need to grow up healthy, caring, and responsible. Roehlkepartain and colleagues (2007) claim that a combination of external assets (support, empowerment, and boundaries) combined with internal assets (positive identities, commitment to learning) is optimal. Youth contribution includes any form of youth engagement, which may include youth involvement in community service and decision-making (Hinson et al., 2016). Finally, an enabling environment in PYD suggests that youth are surrounded by an environment that supports their assets and agency while strengthening their ability to avoid risk and stay safe and secure (Hinson et al., 2016).

Although PYD has shaped youth programming globally, there is a dearth of literature investigating and documenting the impacts of PYD

* Corresponding author.

E-mail address: chow@email.unc.edu (G. Chow).

<https://doi.org/10.1016/j.childyouth.2023.107110>

Received 17 September 2021; Received in revised form 28 June 2023; Accepted 23 July 2023

Available online 25 July 2023

0190-7409/© 2023 Published by Elsevier Ltd.

programs in low- and middle-income countries (LMICs). To date, minimal literature has documented the impacts of PYDs (Alvarado et al., 2017) across the globe. Alvarado and colleagues found a limited number of programs that employed rigorous methods to investigate the impacts of PYD, indicating that gaps remain in understanding the effects of PYD programming. Furthermore, Alvarado et al. (2017) documented that there are fewer PYD programs for vulnerable and marginalized youth (VMY) populations. However, Alvarado et al. (2017) do not exclusively focus on this population, which is where our paper departs from the existing reviews of PYD in LMICs. In this paper, we define VMY as youth who are often excluded from social, economic, or educational opportunities their peers enjoy due to characteristics and conditions beyond the youth's control (Auerswald, Piatt, and Mirzazadeh, 2017). VMY are generally excluded or underserved due to gender, ethnicity, race, (dis)ability, sexual orientation, migration status, economic status, religion, education level, and mental or physical health (Powers et al., 2014). Youth can be further marginalized due to their age. The interplay of age and group exclusion negatively impacts the development and progression of youth (Powers et al., 2014).

1.1. Definition and significance of positive youth development

Positive Youth Development (PYD) emerged from the need for youth development practitioners to use strengths-based prevention approaches in addressing youth issues (Catalano et al., 2002). Although PYD was pioneered by youth development practitioners who developed programs with a prevention focus, theory, and research on this approach followed. PYD is used to engage youth development in a way that focuses on the strengths of the youth. While there are several definitions of PYD, we defined PYD broadly as youth's developmental experiences that provide optimal preparation for adulthood (Lerner, 2000). In this paper, PYD refers to a programmatic approach for supporting healthy, productive, and engaged youth as they grow into adulthood by building their skills, assets, and competencies, fostering healthy relationships and strengthening the environment and systems that serve youth (YouthPower, n.d.). Consistent with YouthPower's PYD approach, PYD programs work with youth to improve their assets, agency, enabling environment, and contribution. Assets are defined as necessary resources and skills to achieve desirable outcomes. Agency refers to youth's ability to use their assets and skills, including social and emotional competencies, to make informed decisions about their lives, set their own goals, and act on those decisions without fear of violence or retribution. An enabling environment provides youth opportunities to enhance their assets and agency and facilitate access to social and economic resources. Contribution refers to youth's involvement and leadership in their communities' efforts to promote PYD.

Research shows that youth who are surrounded by positive encounters and interventions engage in less risky behaviors and successfully transition into adulthood (Alberts et al., 2006; Bandy & Moore, 2010; Eccles & Gootman, 2002; Roth & Brooks-Gunn, 2000; Pittman, Irby, & Ferber, 2001; Lerner, 2004; Lerner, Bowers, Geldhof, Gestsdottir, & DeSouza, 2012; Pittman, 1999; Lerner & Lerner, 2013; Catalano, Berglund, Ryan, Lonczak, & Hawkins, 2004). PYD provides positive opportunities for youth, and these opportunities can, in turn, prevent risky behaviors among young people and lead to improvements in their social and emotional outcomes. In high-income countries (HICs), the documented positive social and emotional outcomes that youth who participate in activities include: attitudes and academic performance and reductions in internalizing symptoms and risky behaviors (Durlak et al., 2011), positive adolescent and reproductive health outcomes (Gavin, Catalano, David-Ferdon, Gloppen, & Markham, 2010), and lower risk of having personal, social, and behavioral problems (Alvarado et al., 2017; Catalano et al., 2019).

Research has constantly indicated that vulnerable and marginalized youth are especially subject to experiencing negative outcomes due to a lack of targeted interventions that address the specific issues that VMY

populations experience (Vojt, Skivington, Sweeting, Campbell, Fenton, & Thomson, 2018). For example, there appears to be a need to develop interventions that provide accommodation for youth with disabilities (Harwick, Tyre, Beisse, & Thomas, 2015); LGBTQ youth who may experience stigma and discrimination (Abreu & Kenny, 2018), family disapproval, and violence, immigrant and refugee youth (Edge, Newbold, & McKeary, 2014) who are likely to experience stress due to societal exclusion, poverty and trauma and young women (Camacho-Miñano, LaVoi, & Barr-Anderson, 2011), particularly in communities with social norms that prevent them from thriving. Evidence of the impacts of such targeted programs may provide critical information for improving programming that will adequately address the nuanced needs of VMYs.

1.2. Study purpose

In another manuscript based on the systematic review of Alvarado et al. (2017), Catalano et al. (2019) provide emerging evidence that PYD programs in LMICs are feasible and have resulted in positive impacts across a range of outcomes, including health and employment. Although this systematic review of efficacy substantially contributed to the review and synthesis of PYD in LMICs, the impacts of PYD programs on VMY populations in LMICs remain understudied. This paper aims to describe and assess the published literature, from 2010 to 2016, on PYD programs for VMY populations in LMICs. First, we identified the different PYD programs implemented specifically for VMY populations in LMICs. Second, we described these programs based on the PYD domain, youth development sector, location, and outcomes. Third, we reviewed each program's research design, emphasizing the quality of evaluation designs. Fourth, we discussed our findings, including implications for future research on PYD programs for VMY in LMICs.

2. Methods

Our scoping review described the state of PYD programming for vulnerable and marginalized youth in LMICs and assessed the quality of available evidence documenting its effects on youth outcomes. Given the heterogeneous and broad scope of our topic, we conducted a scoping review, making it less compatible with a full systematic review (Peters et al., 2015; Munn et al., 2018). Adapting the procedures outlined by Peters et al. (2015) and Khan et al. (2003), our systematic scoping review was conducted using four phases of a systematic review: preparation, retrieval, screening and appraisal, and synthesis. These four phases added rigor to our review and provided a foundation for a subsequent systematic review. Our systematic scoping review included an assessment of quantitative, qualitative, and multimethod studies to provide a holistic review of current knowledge regarding PYD programming for VMY populations (Higgins & Greens, 2012). In our scoping review, we found that quantitative studies identified the impact of PYD programs; qualitative studies described the process, facilitators, and barriers; and multimethod studies combined findings from quantitative and qualitative methods to generate further insights on the phenomenon being studied.

2.1. Preparation

The protocol for this study adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines for systematic reviews (Moher et al., 2009). During Phase 1 (preparation), the research team developed appropriate search terms in consultation with a reference librarian and field experts. Search terms included three broad categories: (1) vulnerability and marginalization, (2) target population, and (3) location. The three broad categories of search terms were linked by the connector "AND" in the search: (1) vulnerability and marginalization (vulnerability, marginalization, underserved, sensitive, stigmatize, stereotype, and discrimination); (2) target population

(youth, adolescent, teen, young, young person, young adult, early adult, and minor) and (3) location (low and middle income, international, Africa, Asia, Southeast Asia, Caribbean west indies, south America, Latin America, Central America, middle east, and all countries classified as low and middle income following the World Bank classification list).

A priori eligibility criteria were developed to guide the inclusion and exclusion of studies. Publications were excluded if they: (i) provided insufficient information (e.g., abstracts only); (ii) were published before 2010 or after 2016; (iii) were not specific to LMICs; (iv) were published in a language other than English; (v) did not refer to the concept of vulnerability or marginalization; (vi) or did not focus on youth populations between the ages of 15–24. For articles related to PYD programs, we used YouthPower Learning's definition of PYD, which aligns with the key elements of the USAID Youth Development in Policy report and existing definitions and frameworks of PYD (e.g., [Alvarado et al. 2017](#); [Catalano et al. 2002](#); [Eccles and Gootman 2002](#); [Lerner 2004](#)). YouthPower Learnings, definition states that PYD engages youth along with their families, communities, and government so that youth are empowered to reach their full potential. PYD approaches build skills, assets, and competencies; foster healthy relationships; strengthen the environment; and transform systems (YouthPower, n.d.).

In addition to these criteria, to be identified as a PYD program, the program had to integrate at least one of the four PYD domains (assets, agency, contribution, and enabling environment) into its programming for VMY populations.

2.2. Retrieval

Our research review included peer-reviewed and grey literature or research produced outside of traditional academic mediums (i.e., refereed journal articles). Grey literature refers to organizational documents, government reports, or white papers. The research team searched 19 databases and collections, including those maintained by international development organizations and donors.

We used Inter City Fund's (ICF) Document Classification and Topic Extraction Resource (DoCTER), an automated document classification web-based tool (Varghese, Cawley & Hong, 2018). DoCTER greatly reduced manual effort in identifying 20,761 records following a comprehensive search process, which tends to result in low search precision ([Haynes et al., 1994](#)) due to the wide breadth of subject areas retrieved. DoCTER uses machine learning techniques to automate document classification. A review of machine learning-based text classification to identify studies for systematic reviews suggested that machine learning may be used to replace the traditional secondary reviewer in systematic reviews ([O'Mara-Eves et al. 2015](#)). Consistent with suggestions in the literature (Varghese et al., 2018), we utilized a supervised clustering approach to classify the 20,761 records. Forty citations were randomly selected as seed studies from the larger collection of unclassified documents. The random selection of seed documents ensures that the seeds do not influence cluster formation more than the other unlabeled documents. The seeds are used to supervise the labeling process by which entire clusters are labeled as relevant or not (Varghese et al., 2018). Each citation received a numerical priority code ranging from 0 (least relevant) to 6 (most relevant). We selected records assigned a priority code of 4, 5, and 6 ($n = 7,036$) for inclusion in the screening phase. We also randomly checked 10–15 citations from the lower priority codes (0–3) to verify the accuracy of priority codes. Overall, the assignment of the priority codes based on title and abstract relevance was accurate.

2.3. Assessment and synthesis

The review team of four researchers, screened the remaining 7,036 titles and abstracts in Covidence, an online systematic review management system. First, two researchers independently reviewed titles and made a unanimous decision as to which abstracts would be included in

the next round of screening. Second, pairs of researchers independently reviewed abstracts. If a consensus was not reached, a third researcher was consulted—the full texts of all the eligible articles remaining after abstract screening were obtained. Two pairs of researchers independently assessed the methodological quality of included studies. We descriptively summarized the data by documenting each program's evaluation design, research methods, and outcomes. More specifically, we utilized an Excel spreadsheet to categorize the full-text articles into our main topic areas (e.g. Youth Experiences, PYD, PYD Barriers, PYD Facilitators, Scaled Programs, and Social Determinants). Findings from the full-text articles were summarized within each topic area to determine emergent themes.

3. Results

As depicted in [Fig. 1](#), 24,373 articles were recovered using the initial search terms. Only 20,761 records remained after scanning and removing duplicates. To identify the most relevant citations, supervised clustering in DoCTER was conducted, and 7,036 records remained. The title and abstracts of these records were screened using a priori criteria, resulting in 1,019 articles. We added another layer of inclusion criterion after the title and abstract stage, which were specific to this review, i.e., publications must be about PYD programs for VMY in LMICs. Thirty-four peer-reviewed and eight grey citations were included in the final review—the 42 citations described 38 unique programs identified as PYD.

3.1. PYD programming and characteristics

The PYD framework depicts an interplay between its four domains: Assets, Agency, Contribution, and Enabling Environment. Consistent with the results of an earlier YouthPower Learning systematic review on PYD programs in LMICs ([Alvarado et al., 2017](#)), most programs in our review did not identify using a PYD approach. One exception was Compassion International's youth development projects in El Salvador ([Tirrell et al., 2019](#)). For our review, we classified programs as using a PYD framework if at least one of the four PYD domains was integrated into programming for vulnerable and marginalized youth (VMY) populations. Using this definition, we identified 38 programs in our review. We used the Positive Youth Development Measurement Toolkit ([Hinson et al., 2016](#)) to identify whether a program included one of the four PYD domains. In particular, we used the Positive Youth Development Illustrative Indicators as a checklist for identifying whether each program for VMY aligned with the PYD framework, its four domains, constructs, definitions, and indicators (see [Hinson et al., 2016](#)). For example, a program that provided training in skills specific to vocation, employment, or financial capacity was classified as using the asset domain.

Similarly, programs were classified as consistent with PYD's contribution domain if program descriptions included meaningful youth engagement through advocacy, volunteering, youth-focused clubs, and peer mentorship. Programs were considered to align with PYD's agency domain if there was a reported increase in positive identity or self-efficacy. Finally, programs that reported improved bonding with family members or school personnel and an increase in prosocial norms. [Table 1](#) summarizes the 38 PYD programs for VMY in LMICs that we identified in our literature review. The table alphabetically organizes the citations by country and world region.

3.2. Programming by PYD domains

Thirty-five of the 38 programs for VMY had an *asset development* component, which comprises skills training, exposure to formal education, interpersonal skills, higher-order thinking skills, emotion recognition, self-control, and academic achievement. In addition to these intangible *assets*, cash transfers were a common financial asset component. The second most common PYD domain represented in VMY

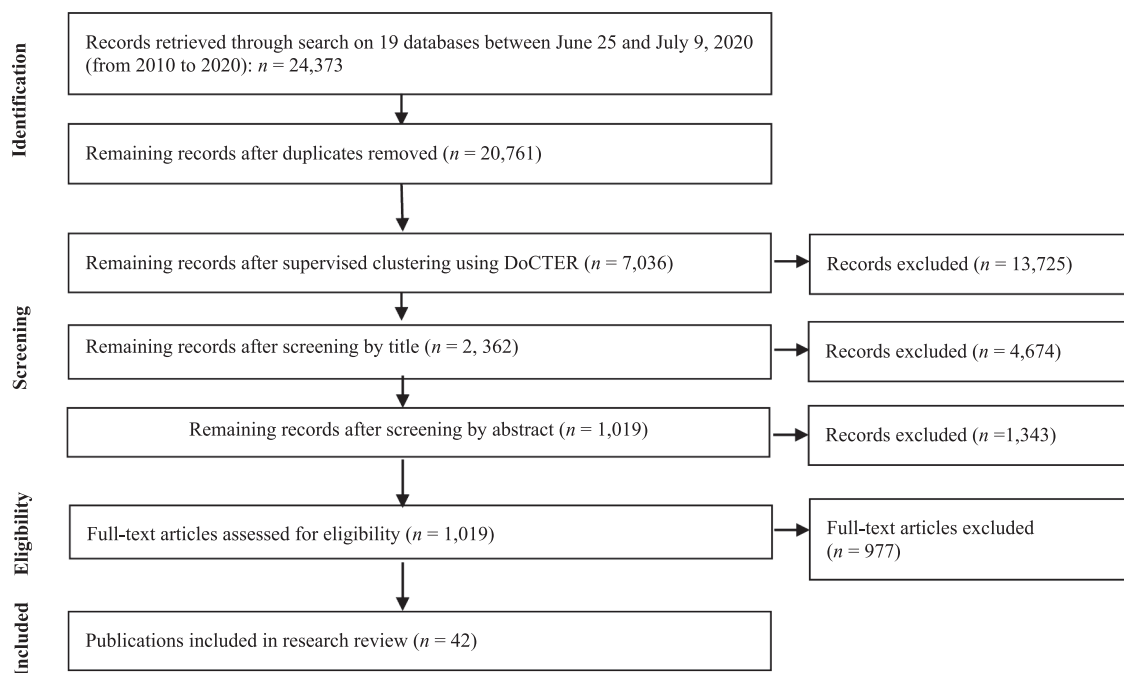


Fig. 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram of screening and selection.

programming was *enabling environment*. *Enabling environments* include the social, normative, structural (e.g., laws, policies, programs, and systems), and physical environment (e.g., safe, supportive spaces environment) that maximize youth's ability to avoid risks, stay safe, and secure, and be protected. The prevalence of programs for VMY addressing the enabling environment PYD domain illustrates the precarious conditions of VMY, many of whom attended programs because they offered a safe, supportive space without fear of social stigma (Embleton et al., 2019; Váľková et al. 2020). In addition to assets and enabling environment, over a third of PYD programs addressed the *agency* domain. *Agency* facilitates the development or enhancement of youth's positive identity, self-efficacy, perseverance, positive beliefs about the future, and ability to plan. *Contribution* was the least common PYD domain addressed in the 38 programs we reviewed. This low number reflects the limited opportunities for VMY to participate fully in various activities and meaningfully engage in an inclusive and mutually respectful partnership in which VMY are viewed as experts on their issues and needs.

Three programs included all four PYD domains (Tirrell et al., 2019; Stackpool-Moore et al., 2016; van Esbroeck et al., 2016). We also identified programs that included three PYD domains, with the most common pattern being assets, agency, and enabling environment ($n = 7$ programs), followed by assets, agency, and contribution ($n = 3$), and assets, contribution, and enabling environment ($n = 2$). Thirteen programs covered two of four PYD domains, with assets and enabling environment ($n = 10$) being the most common, followed by assets and agency ($n = 2$) and agency and enabling environment ($n = 1$). Ten programs included one PYD domain, with seven of these programs focusing on assets, two on enabling environment, and one on contribution. See Table 1 for details of PYD domains.

3.3. Programming by youth development sector

Programs for VMY populations were distributed almost uniformly across different sectors as depicted in Fig. 2. We used Alvarado et al.'s (2017) classification of programs into three primary sectors: (a) Health, (b) Democracy and Governance, and (c) Economic Development and Education. Health was the most commonly represented program sector ($n = 20$), with programs related to Democracy and Governance ($n = 18$)

and Economic Development and Education ($n = 16$) represented at comparable levels. Among the 38 programs we reviewed; 13 addressed more than one sector. Four programs combined Health with Democracy and Governance topics; four integrated Health with Economic Development and Education; two combined Democracy and Governance and Economic Development and Education; and three programs addressed all three sectors. The three programs with all three sectors as foci were the Adolescent Girls Empowerment Program (Zambia), the Adolescent Girls Initiative (Kenya), and the Marriage: No Child's Play program (India, Malawi, Mali, Niger, and Pakistan). All three programs covered topics such as family planning and reproductive health, child marriage, violence, education, and economic resources, primarily focusing on adolescent girls.

3.4. Programming by region and country

Our review identified programs for VMY in 25 LMICs. Sub-Saharan Africa (SSA) was the most represented region, with 12 countries and 24 programs. South Asia was second, with five countries and 12 programs. Six programs were established in four Latin America and the Caribbean countries, three in Southeast Asia, two in the Middle East and Eurasia, and one in Oceania/Pacific Islands. Among the 38 reviewed programs, three were implemented in multiple LMICs, such as the Marriage: No Child's Play program by More than Bride Alliance (India, Malawi, Mali, Niger, and Pakistan), My Rights, My Voice (Afghanistan, Georgia, Mali, Nepal, Niger, and Pakistan), and the Link Up Project (Bangladesh, Ethiopia, Myanmar, and Uganda). India ($n = 6$) had the greatest number of programs, followed by Kenya ($n = 4$), Ethiopia, Tanzania, and Uganda ($n = 3$), and Brazil ($n = 3$).

PYD Outcomes. We classified PYD outcomes using the socio-ecological model of youth development (Atkiss et al., 2011). The classification included individual, institution, or community, and structural impact of PYD and other youth-focused programming on VMY's well-being. Our review of the 38 programs revealed that individual-level outcomes remained the most common outcome evaluated ($n = 38$). Community- and institutional-level outcomes were assessed in eight programs, whereas five programs examined structural-level outcomes.

Individual outcomes: Self-efficacy was one of the most frequently evaluated individual outcomes. Researchers observed a positive impact

Table 1
A Summary of Positive Youth Development Programs for VMY in Low- and Middle-Income Countries.

Citations	Location	World Region	PYD Domain	Program Target Population	Sector	Program	Type of Evaluation	Research Method	Evaluation Outcomes (examples)
Mat, MLJ Le	Ethiopia	Africa	AS	In- and out-of-school youth	EDE	Sexuality education	Process	Qualitative	Mechanisms through which gender, power and sex influence exclusionary practices
Embelton et al 2019	Kenya	Africa	AS, AG EE	Street-connected youth (16 to 24 years old)	H, EDE	Stepping Stones ya Mshefa & Kujijenga Kimaiisha: adaptation of the combined Stepping Stones and Creating Futures interventions	Process	Mixed	Acceptability and feasibility of adapting existing evidence-informed interventions
Grammaans et al 2019	Kenya	Africa	AG, EE	PYD barriers to program adaptation	H, DG	Anti FGM/C	Process	N/S	Barriers to participation
Austrian et al 2020	Kenya	Africa	AS, AG, EE	Adolescent girls 11–15 in two marginalized areas of Kenya	H, EDE, DG	Adolescent Girls Initiative	Impact & Economic	Quantitative, Experimental, Cluster randomization	Health (e.g., sexual and reproductive health knowledge), education (e.g., transition to secondary school), cost-effectiveness
Sikenyi 2017	Kenya	Africa	AS, C, EE	Young people ages 18–35	EDE	Youth Enterprise Development Fund	Process	N/S	Implementation challenges, including youth outreach
Kaunda-Khangamwa, et al 2019	Malawi	Africa	AS, EE	Adolescents living with HIV (15–19 years old)	H	Teen health club	N/S	Mixed, case studies	Social lives and health needs of ALHIV,
Burke et al 2019	Mozambique	Africa	AS, AG, EE	13–19 years old, adolescent girls.	EDE, DG	Women First (part of World Vision's Strengthening Communities through Integrated Programming)	Impact	Quantitative, Quasi-experimental	Knowledge of gender-based violence and school attendance
Uduji et al 2019	Nigeria	Africa	C	Rural youth	EDE	Cultural tourism projects as economic interventions	Process	N/S	Inclusion/participation of youth
Anosike 2019	Nigeria	Africa	AS, AG, C	Youth in conflict areas	EDE	Human capital and entrepreneurship education	Process	Qualitative	Program adaptation to meet needs of VMY
Chau 2016	Senegal	Africa	AS, AG EE	Primary and secondary students	H	Sexuality education	Process	Mixed	Enabling factors and barriers to scale up
Forbes-Grenade 2019	South Africa	Africa	AS, AG, C	Girls	DG	Girls in Risk Reduction Leadership program	N/S	N/S	Strategies for inclusion of disadvantaged adolescent girls in decision-making and action related to the reduction of vulnerability and risk.
Sitienei 2019	South Africa	Africa	AS, EE	Orphans and vulnerable youth	H, EDE	Psychoeducation and social intervention	Impact	Qualitative	Psycho-educational and social well-being
Iseselo et al 2019	Tanzania	Africa	AS	18–35 years old, unemployed without a reliable source of income, living in urban camps.	H, EDE	Entrepreneurial skills training	Impact	Qualitative	Social capital, financial management
Kail et al 2020	Tanzania	Africa	AS	Youth affected by HIV/AIDS and their caregivers	DG	Innovative model of family care/ Better Future International model	Impact	Qualitative	Social capital
Ssewamala 2018; Nabunya 2019; Tozan 2019	Uganda	Africa	AS, AG EE	AIDS-affected adolescents in Uganda	H, EDE	Bridges or Bridges PLUS, family economic intervention with matched savings accounts	Impact	Quantitative, experimental design	Health, mental health, self-concept, self-efficacy, HIV knowledge
Adelman et al 2019	Uganda	Africa	AS	Girls living in camps for internally displaced persons	H	School feeding program and take-home ration	Impact	Quantitative; Experimental, Cluster randomization	Anemia

(continued on next page)

Table 1 (continued)

Citations	Location	World Region	PYD Domain	Program Target Population	Sector	Program	Type of Evaluation	Research Method	Evaluation Outcomes (examples)
Chirwa-Kambole 2020	Zambia	Africa	AS, EE	Adolescent girls between the ages of 13 and 18 years old.	H, DG	Comprehensive Sexual and Reproductive Health Education (CSRHE) youth clubs	Process	Qualitative	Acceptability of CSRHE youth clubs
Austrian et al 2020 ; Austrian et al 2018	Zambia	Africa	AS, EE	Adolescent girls (10–19), never married	H, DG, EDE	Adolescent Girls Empowerment Program	Impact	Quantitative, Experimental, Cluster randomization	Sexual and reproductive health knowledge, gender norms, education, risk behaviors, self-efficacy, and financial literacy
Freudberg 2018	India	Asia	AS, C, EE	Adolescent boys (14–17) and young men (18–25)	H, DG	Community-level gender equality intervention	Impact	Mixed	Knowledge and attitudes related to gender, sexuality, and violence.
Ayubi 2018	Afghanistan	Asia	EE	Youth who need counseling services	H	Family counseling	N/S	N/S	N/S
Huda 2018	Bangladesh	Asia	AS, EE	Poor, slum-dwelling girls, especially married adolescent girls	H	Strengthening Evidence for Programming on Unintended Pregnancy (STEP UP)	Process	N/S	Process for expanding access to family planning services for married adolescent girls
Vyas 2020	India	Asia	EE	Adolescent girls and young women	DG	Gender equality campaign	Process	Mixed	Reach, engagement, and perceptions of the campaign messaging
Nagler 2019	India	Asia	AS, EE	Youth from low-income families	H	School-based tobacco prevention program	Impact	Qualitative	Agency, optimism, confidence, communication skills, future orientation
Gonsalves et al 2019	India	Asia	AS	Adolescents with or at risk for mental health issues	H	Problems – Options – Do-It (POD) Adventures	Process	Mixed	Youth participation in development of a mobile-app intervention
Mathias 2019	India	Asia	AS, AG	Youth with psycho-social disability	H, DG	Psychosocial intervention to improve mental health	Impact	Qualitative	Social inclusion, mental health
Lopus et al 2019	Indonesia	Asia	AS	Low-income youth	EDE	Financial education	Impact	N/S	Financial literacy knowledge, soft skills
Bilagher et al 2020	Iraq	Asia	AS, AG, EE	Out-of-school, disenfranchised children aged 12–18 years, particularly adolescent girls	EDE	Accelerate Learning Program	Process	N/S	Scaling up of accelerate learning programs
Zipp 2017	St. Lucia	Caribbean	AS, EE	At risk girls	DG	Sports development program	Impact	Qualitative	Capability development (e.g., self-efficacy, challenging gender norms, peer and mentor relationships, and body image)
Tirrell et al 2019	El Salvador	Central America	AS, AG, C, EE	Low-income youth	DG, EDE	Compassion International Positive Youth Development (PYD) -informed programming	Impact	Quantitative, quasi-experimental	Spirituality, character, connection
Sosa-Rubi 2017	Mexico	North America	AS, EE	Youth in urban, low-income high schools in Mexico City	DG	True Love, a school-based program to reduce dating violence	Impact	Quantitative (quasi-experimental design)	Dating violence behaviors, Beliefs related to gender norms, Knowledge, and skills for preventing dating violence
Heard et al 2019	Samoa	Oceania	AS, AG, C	Youth	DG	Intimate partner violence prevention using interactive drama	Process	Qualitative	Understandings of intimate partner violence, intimate partner relationships and cultural gender role expectations
Figueiredo et al 2019	Brazil	South America	AS, AG	Young people in favelas	EDE	Social intervention combined with arts	Impact	Mixed	Personality and character development, communication skills, social skills, and personal motivation.
Borloti 2017	Brazil	South America	AS	N/S	H	School-based drug prevention programs	Process	N/S	Quality of school-based drug prevention

(continued on next page)

Table 1 (continued)

Citations	Location	World Region	PYD Domain	Program Target Population	Sector	Program	Type of Evaluation	Research Method	Evaluation Outcomes (examples)
Valkova 2020	Brazil	South America	AS, EE	LGBT youth (17 to 23 years old)	DG	Sport for Development and Peace	Process	Qualitative	Implementation of sexuality-focused programs for LGBT youth
Van Esbroeck et al 2016	Afghanistan, Georgia, Mali, Nepal, Niger and Pakistan, Tanzania, and Vietnam	Spans World Regions	AS, AG, C, EE	N/S	EDE, DG	My Rights, My Voice	Process	Mixed	Youth participation/ inclusion
Stackpool-Moore	Bangladesh, Myanmar, Uganda and Ethiopia	Spans World Regions	AS, AG, C, EE	Young men who have sex with men, young sex workers, young people who use drugs, young transgender people, young homeless people, and other vulnerable young people at risk for HIV transmission	H	Link Up: Access to integrated HIV services and sexual and reproductive health and rights (SRHR) for marginalized young people	Process	N/S	Barriers and facilitators of access to Sexual and Reproductive Health Rights (SRHR)
Pop Council 2017; 2020	India, Malawi, Mali, Niger and Pakistan	Spans World Regions	AS, AG, EE	Adolescent girls 12 to 19	H, DG, EDE	Marriage: No Child's Play (More Than Brides Alliance)	Impact	Mixed	Sexual and reproductive health knowledge, gender equality, education
Mat, MLJ Le	Ethiopia	Africa	AS	In- and out-of-school youth	EDE	Sexuality education	Process	Qualitative	Mechanisms through which gender, power and sex influence exclusionary practices

Note: AS = Assets, AG = Agency, C = Contribution, EE = Enabling Environment; H = Health, DG = Democracy and Governance; EDE = Economic Development and Education; N/S = Not specified.

on youth's self-efficacy in several asset development programs for adolescent girls and AIDS-affected youth (Austrian *et al.* 2020a; Austrian *et al.* 2020b; Ssewamala *et al.* 2018; Tozan *et al.* 2019) in a social inclusion and mental health program for youth with disabilities (Mathias *et al.* 2019), and in a sports development program for young survivors of interpersonal violence (Zipp 2019). Knowledge was another commonly evaluated individual outcome. A positive effect on knowledge related to sexual and reproductive health and right (SRHR), HIV, financial literacy, and child marriage was reported in various asset development and financial education programs (Austrian *et al.* 2020a; Austrian *et al.* 2020b; Lopus *et al.* 2019; Melnikas *et al.* 2019; Ssewamala *et al.* 2018; Tozan *et al.* 2019). In one, Tozan *et al.* (2019) found that a positive impact on self-efficacy and knowledge was observed at two- and four-year follow-ups post-intervention, which indicates that programming for VMY can sustain positive effects years after a program ends. Evaluations of VMY programming also documented positive effects on gender-biased attitudes (Sosa-Rubi *et al.* 2017), self-concept (Ssewamala *et al.* 2018), optimism (Nagler *et al.* 2019), character (Figueriedo *et al.* 2019; Tirrell *et al.* 2019), spirituality (Tirrell *et al.* 2019), psychological well-being (Mathias *et al.* 2019; Sitienei *et al.* 2019), soft and entrepreneurial skills (Figueriedo *et al.* 2019; Iseselo *et al.* 2019; Sitienei *et al.* 2019), financial management skills (Iseselo *et al.* 2019); and individual wealth (Austrian *et al.* 2020b). Reductions in transactional sex among vulnerable adolescent girls were also observed in the AGEV (Austrian *et al.* 2020a).

The key desired outcomes (e.g., higher-order thinking skills, self-control, positive identity, self-efficacy, and positive beliefs about the future) of PYD programs were not consistently measured and evaluated across the programs we reviewed. Still, several programs reported positive effects on interpersonal skills, financial capacity, formal education, academic achievement, self-efficacy, optimism, positive identity, prosocial norms, recognition by adults, and opportunities for prosocial involvement (Austrian *et al.* 2020a; Figueriedo *et al.* 2019; Mathias *et al.* 2019; Melnikas *et al.* 2019; Nagler *et al.* 2019; Sitienei *et al.* 2019; Ssewamala *et al.* 2018; Tirrell *et al.* 2019; Tozan *et al.* 2019; Zipp 2019). However, only the documented effects on self-efficacy, identity, and financial capacity came from programs that used experimental designs that could provide strong evidence of causality (Austrian *et al.* 2020a; Ssewamala *et al.* 2018; Tozan *et al.* 2019).

Institutional and Community Outcomes. Institutional and community outcomes are defined as those outcomes measured at the institutional and community level, e.g., youth-friendly policies and services. However, in this review, findings indicate that these outcomes were not measured. Instead some outcomes are more reflective of aggregated individual outcomes at the community and institutional level. For this reason, we report these outcomes as part of institutional and community outcomes. Among the studies reviewed, outcomes related to schools found positive impacts on school enrollment, attendance, retention, academic achievement (Adelman *et al.* 2019; Melnikas *et al.* 2019; Ssewamala *et al.* 2018; Nabunya *et al.* 2019) and reducing violence in schools (Sosa-Rubi *et al.* 2017). Beyond schools, these programs have yielded positive effects on a broad scope of contextual outcomes, including community participation and recognition of youth's economic and entrepreneurial abilities (Iseselo *et al.* 2019; Mathias *et al.* 2019); peer and mentor relationships (Zipp 2019); social capital (Kail *et al.* 2020); perceived social inclusion (Mathias *et al.* 2019); reduction in the prevalence of psychological and sexual violence (Sosi-Rubi *et al.* 2017).

Structural outcomes. Similar to the definition of institutional and community outcomes, structural outcomes include economic opportunities and social norms. We report what studies in the review presented as outcomes. To rigorously evaluate programs (i.e., AGEV in Zambia and COMPASS in Ethiopia) did not find evidence of an effect on participants' economic and educational outcomes (Austrian *et al.* 2020a; Melnikas *et al.* 2019; Stark *et al.* 2018). Also, AGEV in Zambia did not have an impact on fertility outcomes (Austrian *et al.* 2020a). Similarly, Women First, an economic and social empowerment program for adolescent girls

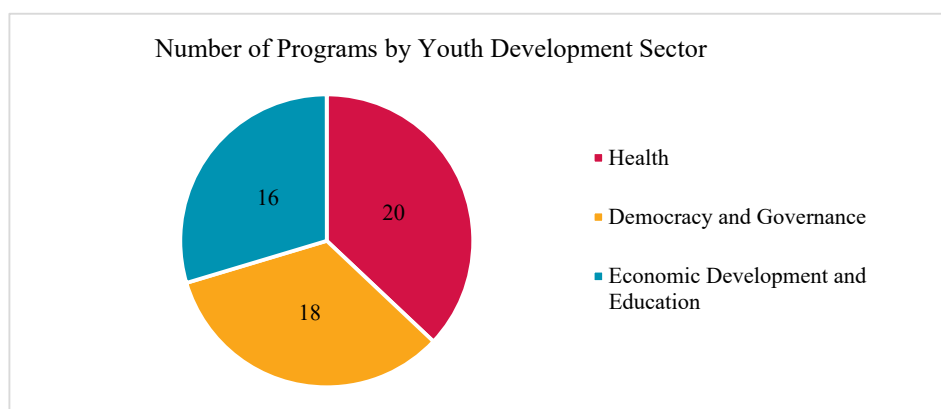


Fig. 2. Number of Programs by Youth Development Sector.

and young women in Mozambique, did not have an impact on knowledge of gender-based violence or school attendance (Burke et al. 2019). Although most programs did not assess a structural outcome, the few programs found no impact on structural outcomes, such as norms related to gender equity (Austrian et al. 2020) and collective self-efficacy (Figueriedo et al. 2019).

3.5. PYD outcomes by development sectors.

Our review revealed variations in the direction of outcomes across three development sectors: health, democracy and governance, and economic development and education. First, PYD programs appear to have a consistent positive impact on knowledge and attitudes across development sectors – health, democracy and governance, and economic development and education (e.g., Austrian et al. 2020a; Austrian et al. 2020b; Lopus et al. 2019; Melnikas et al. 2019; Ssewamala et al. 2018). Second, PYD programs for VMY also had a positive impact on outcomes in the economic and education sector, including school enrollment and academic achievement (Adelman et al. 2019; Melnikas et al. 2019; Ssewamala et al. 2018; Nabunya et al. 2019) and economic and entrepreneurial skills (Iseselo et al. 2019; Mathias et al. 2019; Sitienei et al. 2019). Third, PYD programs' positive impact on outcomes in the democracy and governance sector include better interpersonal relationships, which improve interactions between citizens to ensure that all voices are heard (Zipp 2019); higher social capital, which improves the representation of groups represented at governing bodies (Kali et al. 2020); perceived social inclusion (Mathias et al. 2019); and reduction in the prevalence of school-based violence (Sosi-Rubi et al. 2017). Although these results are promising, the limited number of PYD programs for VMY does not allow for a conclusive determination of impact across development sectors.

3.6. Study Characteristics: Quality of evaluation design

Few programs have been rigorously evaluated using experimental and quasi-experimental designs, as most programs either have not been evaluated or have been evaluated using designs that can only provide weak evidence of causality. Of the 38 programs, five were evaluated using an experimental design. These programs are listed in the next paragraph. Another five programs were evaluated using a quasi-experimental design (Burke et al., 2019; Saul et al., 2017; Sosi-Rubi et al., 2017; Tirrell et al., 2019; van Esbroeck et al., 2016). The remaining 28 programs were either evaluated using nonexperimental designs or were not evaluated. Also, we did not find rigorous published research identifying the specific program features responsible for impact. Rather, the published studies that we reviewed remarked on which program features could be responsible for impact, but these remarks were anecdotal and did not emerge from the systematic

collection, analysis, and interpretation of data and evidence.

Experimental and quasi-experimental designs. We identified five programs that had been evaluated using an experimental design: Bridges/Bridges Plus in Uganda (Ssewamala et al. 2018); Adolescent Girls Empowerment Program (AGEP) in Zambia (Austrian et al. 2020a); Adolescent Girls Initiative – Kenya (Austrian et al. 2020b); Creating Opportunities through Mentoring, Parental, and Safe Spaces (COMPASS) in Ethiopia (Stark et al. 2018); and the Food-for-Education school feeding program in Uganda (Adelman et al. 2019). Evaluations of the AGEP program in Zambia and the school feeding program in Uganda used a cluster randomized controlled trial design. Although randomization was not a feature of their evaluation designs, three programs—True Love in Mexico (Sosa-Rubi et al. 2017), Compassion International PYD programming in El Salvador (Tirrell et al. 2019), and the Women First program in Mozambique (Burke et al. 2019)—used a quasi-experimental design and advanced quantitative methods to assess program impact.

4. Discussion

There exists limited peer-reviewed literature on PYD programming in LMICs, therefore, there is limited empirical evidence that identifies interventions that are most responsive to VMY's needs. The majority of PYD programs are conducted by international development agencies such as USAID and UNICEF. This review identified more studies that examined the impact of interventions on different developmental outcomes for VMY than studies that assessed the fidelity of intervention receipt. In our discussion, we interpret our findings by demonstrating how our findings can further enhance targeted PYD programs to respond to the needs of VMY optimally.

4.1. PYDs enabling environment

Our review revealed several features of PYD and other youth-focused programming for VMY that influenced the programs' effectiveness and responsiveness to VMY's needs.

In the PYD literature, an enabling environment was strongly affiliated with program efficacy. An enabling environment can include social, structural, and physical environments that support youth while promoting their social and emotional competence to thrive. For instance, VMY need youth-friendly laws and policies related to employment to create jobs that young people can obtain. Employment-related interventions for improving youth's capabilities, aspirations, and efforts will likely be ineffective if youth have limited job opportunities. In addition to laws and policies, youth-responsive services are crucial to VMY, who often lack the social networks, employability and entrepreneurial skills, and business environment to find a job or grow their business. For example, findings from Kenya's Youth Enterprise

Development Fund noted several obstacles to initiating entrepreneurial activities among VMY, including insufficient start-up capital and a lack of accessible places to situate their enterprises (Sikenyei 2017).

In addition to youth-friendly laws and policies and youth-responsive services, gender-transformative approaches are common components of programming for VMY. These approaches seek to improve the precarious circumstances of adolescent girls and young women. Barriers to access to and engagement with many youth-focused programs stem from gender-biased norms and practices that deny girls and young women opportunities to gain new knowledge, attend training sessions, form new relationships with peers, and exercise their agency through improved self-efficacy, hope, or optimism about their future potential, goals, choices, or plans. Fifteen of the 38 reviewed programs, including the AGEF, Adolescent Girls Initiative, COMPASS, and MTBA, targeted adolescent girls and young women. Additionally, an analysis of a youth-focused program in India found that adolescent girls and young women reported the most benefits (Mathias et al., 2019). Thus, gender-responsive policies and services should be included in PYD programming to foster gender equality and promote the participation and involvement of girls and young women. For example, gender-responsive services should integrate youth-identified gender issues into every programming stage (e.g., site selection, staff selection, program development, and content).

An enabling environment for VMY requires opportunities for prosocial involvement, bonding, support, and spaces for their physical and psychological safety. Mentorship is one of the most common components of successful youth-focused programming that has assisted VMY in various life transitions, including securing employment or obtaining further education (Nagler et al. 2019; Zipp 2019). Several studies cited mentorship as a program characteristic linked to higher levels of self-efficacy and better psychological well-being (Siteinei et al. 2019; Zipp 2019). Mentorship may also support VMY in building prosocial behaviors and other desirable intrinsic traits.

Accessible, safe spaces constitute another enabling environment linked to positive outcomes (Nagler et al. 2019; Valkova 2020). Safe spaces offer VMY an environment that fosters a sense of belonging, value, and recognition of youth's strengths and weaknesses (Ahmimed & Quesada-Montano 2019). In India, safe spaces and mentorship opportunities contributed to VMY's sense of belonging and recognition of their positive contributions to community, which influenced their attitudes, behaviors, and future outlook (Nagler et al. 2019). Safe spaces are particularly important to LGBT youth in places where their sexual orientation and gender identity are criminalized. A safe space created for LGBT youth allows them to disclose, claim, and express their sexual identities freely and without fear of intimidation, bullying, or violence (Valkova 2020). Safe spaces can also ensure stigma-free services and programming for VMY, which can increase service use and sustain engagement.

Programs for VMY are more likely to be effective when supported by the youths' families, peer groups, schools, or communities. The emotional attachment and positive interactions between youth and their families, peers, schools, or communities may foster increased confidence and self-efficacy among VMY. For instance, in one intervention in India, young people with psychosocial disabilities and supportive parents experienced the greatest improvements in their self-perceived social inclusion and understanding of mental health issues, which promoted their self-management of their conditions (Mathias et al. 2019).

4.2. Promising programs responsive to VMY's needs

Our review found that multi-component, cross-sectoral approaches offered a promising strategy for developing interventions for VMY. An important component of cross-sectoral approaches is purposeful programming at the community or structural level (Melinkas et al. 2019; Stark et al. 2018). Because many VMY think and behave in ways consistent with social and cultural norms, increasing societal awareness

of and destigmatizing VMY's social circumstances is essential to building effective interventions in a given context. Optimizing interventions' effectiveness at the individual level requires broader structural or cultural transformations. Community sensitization to VMY issues (e.g., adolescent sexual and reproductive health, gender-biased norms, mental health, disability, physical and psychological safety, and sexual identity) requires structural intervention. This intervention should be consistent with the values of the community and meaningfully engage youth in its development and implementation.

An example of a community sensitization program is Suiga/Change, an interactive drama for intimate relationship education among Samoan youth. Suiga/Change uses participatory education to spark discussions of Samoan cultural and social norms related to conflict, violence, and inequality within intimate relationships. As part of the intervention, the audience of the interactive performance participates in an activity exploring the characters and storyline depicted in the drama. In this activity, audience members create sculptures representing the range of motivations and emotions a particular character may be experiencing at a given moment. This activity also provides an occasion for discussing intimate partner violence and its social and cultural influences (Heard et al. 2019). A qualitative evaluation of the intervention found that it increased youth's awareness of intimate partner violence and their motivation to translate this new knowledge into action (Heard et al. 2019).

Cross-sectoral approaches, such as the MTBA and Adolescent Girls Initiative – Kenya, combine training on health, democracy and governance, and economic development and education. These approaches recognize the various asset-related needs of youth and their families, including training in skills specific to vocation, employment, and financial capacity. Cross-sectoral approaches address all PYD domains (i.e., assets, contribution, agency, and enabling environment), recognizing that VMY's needs are multidimensional and complex. One such program is employing a multilevel approach to meet needs and address issues affecting VMY by focusing on education and school curriculum development, male involvement, new stylizations of love and relationships, monitoring and evaluation of the program, and inclusive aspects of religion (Gramaans et al. 2019). This ambitious strategy combines individual- and community-level interventions that address individual needs (e.g., knowledge and information) and community or societal factors (e.g., social norms and stigma) shaping the vulnerability and marginalization of youth.

Cross-sectoral approaches also recognize youth's resilience. This recognition is exemplified by the approaches' focus on asset development, which can involve health, social, or economic assets. Recognizing VMY's resilience may enhance their intrinsic capabilities, including their agency and related outcomes (e.g., positive identity, self-efficacy, ability to plan, perseverance, and positive beliefs about the future), promoting program success.

Because cross-sectoral approaches bridge individual, community, and structural interventions to address VMY's needs, these approaches may require longer-duration programs. Furthermore, because the changes and outcomes of interest (e.g., social norms, transitions to higher education, and labor market) take substantial time, programs should plan for implementation periods of at least five years (Austrian et al. 2020a). This recommendation particularly applies to programs implemented in politically fragile countries or countries with low levels of institutional development (Bilagher et al. 2020; van Esbroeck et al. 2016). Programs in these countries should account for potential delays due to political instability or a need for preliminary infrastructural work.

4.3. Program characteristics for best reach

Our review of the 38 youth-focused programs for VMY in LMICs identified four characteristics that may help optimize programs' reach: peer involvement, community participation, integration into existing programs and services, and creative use of technology. We described

several of these characteristics in our discussion of facilitators of access to PYD and youth-focused programming and elaborated on them here while making further recommendations related to training and media strategies.

First, we believe including VMY's peers as educators or training, facilitators will increase program reach. Peer educators use the same language as VMYs, understand and answer questions clearly, convey information in an engaging and less stigmatizing way than older adults, and talk openly about and more easily relate to sensitive issues (Stackpool-Moore et al. 2017). Peers can also be sources of support, as illustrated by programs implemented in India (Nagler 2019), Malawi (Kaunda-Khangamwa et al. 2010), and South Africa (Forbes-Grenade et al. 2019). Youth peers are also effective recruiters because they belong to the same social networks as VMYs and are likely to be trusted by young people their age and those with similar experiences.

Second, encouraging community members to participate in project implementation can lead to greater outreach, particularly among youth who may not be identified by implementing organizations and their networks (Burke et al., 2019). In addition to leaders and elders, the participation of men in community-based gender equality interventions has a demonstrated potential to increase the involvement of other men (particularly young men) in developing and sustaining social norms that promote gender equality in their communities (Freudberg et al. 2018).

Third, reach may be optimized by integrating programs and services with current community-based youth-focused programming. Health programs for VMY have collaborated with communities' existing service providers in order to maximize their reach and, in some cases, mitigate potential risks for VMY who may use those programs, particularly those related to highly stigmatized issues (Huda et al. 2018; Stackpool-Moore et al. 2017). Utilizing existing institutions and resources is also helpful when scaling up programs. For example, in Bangladesh, the Strengthening Evidence for Programming on Unintended Pregnancy (STEP UP) has the potential to scale up more easily due to its use of existing services such as community health workers and Kazis (or marriage registrars).

Fourth, creative media use has made it more possible to reach many VMY. For example, Girl Rising launched a social media campaign in 2016 in India. The social media campaign (We Dream, We Rise) was designed to stimulate reflections and discussions about adolescent girls and daughters in the country and raise awareness about the gender biases internalized by girls and their families (Vyas 2020). New media, including websites and social media platforms, have also been used to engage a wider audience of VMY in decision-making processes (Hassan 2016). In a recent study, youth reported several media technologies (e. g., hard-copy novels, cell phones, and the Internet) as their primary sources of information about family planning and reproductive health issues (Olumide et al. 2016). Unsurprisingly, the youth's use of media technologies – particularly smartphones – has led to widespread fear and anxiety among older generations that young people, especially adolescent girls, are accessing culturally sensitive content over the Internet (Rao and Lingam 2020). However, this view fails to recognize the potential of Internet access to reach greater numbers of VMY with important information related to their mental and physical health and well-being.

4.4. Variation of needs and Differentiated models of programming for VMY

Recent systematic reviews of interventions designed to improve outcomes of adolescents and young adults (including VMY) in LMICs have shown that while a wide array of interventions can yield desired effects, no one intervention can do so in all contexts and for all outcomes (Kalamar et al. 2016a; Kalamar et al. 2016b; Meinck, 2019). These findings are consistent with our review of PYD and youth-focused programming for VMY. Various interventions focusing on health, economic development and education, and democracy and governance have had demonstrably positive effects on the individual- and community-level

outcomes, while similar interventions did not have any impact on similar individual- and community-level outcomes. The nuanced needs, preferences, and circumstances of different VMY populations cannot be addressed by the same intervention in all contexts and for all outcomes. Rather, an adaptable cross-sectoral approach that addresses the specific needs of VMY populations and their communities will be required to deliver effective programming tailored to different settings. Depending on the desired outcomes, this cross-sectoral approach's components, contents, and materials will likely be different and context-specific. Indeed, our findings underscore the importance of recognizing and engaging with the diversity of VMY. For example, VMY aged 15–18 have different needs and priorities than those aged 19–24. Differentiated programming models will require innovations in service delivery that respond to different contexts and, in turn, reach more VMY and their communities.

Overall, PYD programs are promising interventions to improve the welfare of VMY across health, democracy and governance, economics, and education. However, the current published literature on PYD programs and their impacts remains limited. There is a paucity of PYD programs for other VMY, including youth with disabilities and sexual and gender minority youth, and youth in post-conflict settings. In addition, the reviewed programs have not consistently used comparable measures and examined identical outcomes, which make comparisons across programs in each sector ineffective. The limited number of PYD programs for VMY and the variation of programming components did not allow us to meaningfully compare program effects by type of VMY and age in each development sector. In addition, multicomponent and cross-sectoral programs' cost-effectiveness and benefits valuation remains understudied. Future research should focus on rigorously evaluating PYD program impacts on VMY by development sector, type of vulnerability, and age.

4.5. Limitations

Systematic reviews generally involve a narrow but well-defined research question. Most reviews use the PICO (Population, Intervention, Comparison, and Outcome) framework to formulate a research question. Although systematic literature reviews that use the PICO framework vary in scope, their specific research questions set additional limits or define a narrow set of search terms. The broad scope of the primary research question and the inclusion of multiple secondary questions yielded 24,373 articles. Given the condensed timeline to complete this review, we used an automated document classification technology using machine learning algorithms to categorize each publication according to a fixed set of possible categories. While the ICF's DoCTER improved efficiency in our first round of document classification, it might have eliminated relevant documents (Albalade et al. 2010; Varghese et al. 2017). To minimize this error, two reviewers manually screened the titles of the eliminated documents to ensure the accuracy of the first round of document classification.

4.6. Research recommendations and next steps

First, there is a need to rigorously test multicomponent, cross-sectoral interventions to build evidence of the efficacy of these intervention designs for supporting VMY. Our findings and the youth development field have broadly emphasized the importance of multicomponent, cross-sectoral interventions that address the multi-layering of vulnerabilities among young people. However, little evidence exists regarding best practices and the effects of such interventions. More research and investment in building evidence for multicomponent, cross-sectoral interventions will help establish the relative efficacy of these intervention designs for supporting VMY.

Second, collecting cost data and conducting economic evaluations of multicomponent, cross-sectoral interventions are important steps to identify, measure, value, and compare the costs and results of

multicomponent, cross-sectoral interventions. To date, little evidence supports that cross-sectoral interventions are more cost-effective than individual or standalone interventions when promoting VMY's well-being.

5. Conclusion

Our review found that multicomponent, cross-sectoral approaches offered a promising strategy for developing interventions for VMY. An important component of cross-sectoral approaches is purposeful programming at the community or structural level (Melinkas et al. 2019; Stark et al. 2018). Because many VMY think and behave in ways consistent with social and cultural norms, increasing societal awareness of and destigmatizing VMY's social circumstances is essential to building effective interventions in a specific context. Optimizing interventions' effectiveness at the individual level requires broader structural or cultural transformations. Community sensitization to VMY issues (e.g., adolescent sexual and reproductive health, gender-biased norms, mental health, disability, physical and psychological safety, and sexual identity) requires structural-level intervention. This intervention should be consistent with the values of the community and meaningfully engage youth in its development and implementation.

Human and animal rights

This article does not contain any studies with animals performed by any of the authors.

Informed consent

Informed consent was obtained from all individual participants included in the study.

Ethical approval

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- Abreu, R. L., & Kenny, M. C. (2018). Cyberbullying and LGBTQ youth: A systematic literature review and recommendations for prevention and intervention. *Journal of Child & Adolescent Trauma*, 11(1), 81–97.
- Adelman, S., Gilligan, D. O., Konde-Lule, J., & Alderman, H. (2019). School Feeding Reduces Anemia Prevalence in Adolescent Girls and Other Vulnerable Household Members in a Cluster Randomized Controlled Trial in Uganda. *The Journal of Nutrition*, 149(4), 659–666. <https://doi.org/10.1093/jn/nxy305>
- Alvarado, G., Skinner, M., Plaut, D., Moss, C., Kapungu, C., & Reavley, N. (2017). *A Systematic Review of Positive Youth Development Programs in Low- and Middle-Income Countries*. Washington, DC: YouthPower Learning, Making Cents International.
- Atkiss, K., Moyer, M., Desai, M., & Roland, M. (2011). Positive Youth Development: An integration of the Developmental Assets Theory and the Socioecological Model. *American Journal of Health Education*, 42(3), 171–180.
- Auerswald, C. L., Piatt, A. A., & Mirzazadeh, A. (2017). *Research with disadvantaged, vulnerable and/or marginalized adolescents*. UNICEF.
- Austrian, K., Soler-Hampejsek, E., Behrman, J. R., Digitale, J., Hachonda, N. J., Bweupe, M., et al. (2020). The Impact of the Adolescent Girls Empowerment Program (AGEP) on Short and Long Term Social, Economic, Education and Fertility Outcomes: A Cluster Randomized Controlled Trial in Zambia. *BMC Public Health*, 20(1), 349. <https://doi.org/10.1186/s12889-020-08468-0>
- Bandy, T., & Moore, K. A. (2010). *Assessing self-regulation: A guide for out-of-school time program practitioners*. Washington, DC: Child Trends.
- Benson, P. L., Scales, P. C., Hamilton, S. F., & Sesma, A. (2007). Positive youth development: Theory, research and applications. *Theoretical Models of Human Development*. <https://doi.org/10.1002/9780470147658.chps0116>
- Burke, H. M., Field, S., González-Calvo, L., Eichleay, M. A., & Moon, T. D. (2019). Quasi-Experimental Evaluation Using Confirmatory Procedures: A Case Study of an Economic and Social Empowerment Intervention to Reduce Girls' Vulnerability to HIV in Rural Mozambique. *Evaluation and Program Planning*, 77, Article 101721.
- Camacho-Miñano, M. J., LaVoi, N. M., & Barr-Anderson, D. J. (2011). Interventions to promote physical activity among young and adolescent girls: A systematic review. *Health education research*, 26(6), 1025–1049.
- Catalano, R., Berglund, L., Ryan, J., Lonczak, H., & Hawkins, D. (2002). Positive Youth Development in the United States: Research Findings on Evaluations of Positive Youth Development Programs. *The ANNALS of the American Academy of Political and Social Science*, 591(1), 98–124. <https://doi.org/10.1177/0002716203260102>
- Catalano, R. F., Skinner, M. L., Alvarado, G., Kapungu, C., Reavley, N., Patton, G. C., et al. (2019). Positive Youth Development Programs in Low- and Middle-Income Countries: A Conceptual Framework and Systematic Review of Efficacy. *Journal of Adolescent Health*, 65(1), 15–31. <https://doi.org/10.1016/j.jadohealth.2019.01.024>
- Durlak, J. A., Weissberg, R. P., Dymnicki, A. B., Taylor, R. D., & Schellinger, K. B. (2011). The impact of enhancing students' social and emotional learning: A meta-analysis of school-based universal interventions. *Child Development*, 82(1), 405–432. <http://eric.ed.gov/?id=EJ927868>.
- Eccles, J., & Gootman, J. (2002). *Community Programs to Promote Youth Development*. Washington, DC: National Academy Press.
- Edge, S., Newbold, K. B., & McKeary, M. (2014). Exploring socio-cultural factors that mediate, facilitate, & constrain the health and empowerment of refugee youth. *Social Science & Medicine*, 117, 34–41.
- Embleton, L., Di Ruggiero, E., Okal, E. O., Chan, A. K., Logie, C. H., Ayuku, D., et al. (2019). Adapting an Evidence-Based Gender, Livelihoods, and HIV Prevention Intervention with Street-Connected Young People in Eldoret, Kenya. *Global Public Health*, 14(12), 1703–1717. <https://doi.org/10.1080/17441692.2019.1625940>
- Figueiredo, G., de Oliveira, V., Weilmüller, C., & Orrillo, Y. A. D. (2019). Resultados de Proyectos Socioeducativos En El Desarrollo Humano de Los Jóvenes. *Educación & Realidade*, 44(4), e87645.
- Freudberg, H., Contractor, S., Das, A., Kemp, C. G., Nevin, P. E., Phadiyal, A., et al. (2018). Process and Impact Evaluation of a Community Gender Equality Intervention with Young Men in Rajasthan, India. *Culture, Health & Sexuality*, 20(11), 1214–1229. <https://doi.org/10.1080/13691058.2018.1424351>
- Gavin, L. E., Catalano, R. F., David-Ferdon, C., Gloppen, K. M., & Markham, C. F. (2010). A review of positive youth development programs that promote adolescent sexual and reproductive health. *Journal of Adolescent Health*, 2010(46), S75–S91.
- Harwick, R., Tyre, A., Beisse, K., & Thomas, C. (2015). Intervention targets for youth with disabilities in foster care. *In School Psychology Forum*, 9, 1.
- Haynes, R. B., Wilczynski, N., McKibbon, K. A., Walker, C. J., & Sinclair, J. C. (1994). Developing optimal search strategies for detecting clinically sound studies in MEDLINE. *J Am Med Inform Assoc*, 1, 447–458.
- Heard, E., Fitzgerald, L., Vaai, S., Whittaker, M., Airolupotea, T. J., Collins, F., et al. (2019). Intimate Partner Violence Prevention: Using Interactive Drama for Intimate Relationship Education with Young People in Samoa. *Sex Education*, 19(6), 691–705. <https://doi.org/10.1080/14681811.2019.1597697>
- Hinson, L., Kapungu, C., Jessee, C., Skinner, M., Bardini, M., & Evans-Whipp, T. (2016). *Measuring Positive Youth Development Toolkit: A Guide for Implementers of Youth Programs*. Washington, DC: YouthPower Learning, Making Cents International.
- Iseselo, M. K., Moshia, I. d. H., Killewo, J., Sekei, L. H., & Outwater, A. H. (2019). Can Training Interventions in Entrepreneurship, Beekeeping, and Health Change the Mind-Set of Vulnerable Young Adults Toward Self-Employment? A Qualitative Study from Urban Tanzania. *PLoS One*, 14(8), e0221041.
- Kail, Barbara, Manoj Pardasani, and Robert Chazin. 2020. "The Better Future International's Family Care Model in Tanzania: Creating Social Capital as a Means to Empowerment in Social Work Practice." *International Social Work*, February, 0020872819884986. [10.1177/0020872819884986](https://doi.org/10.1177/0020872819884986).
- Lerner, J.V., Phelps, E., Forman, Y., & Bowers, E.P. (2009). Positive youth development. In R.M. Lerner, & L. Steinberg (Eds.), *Handbook of Adolescent Psychology: Vol. 1. Individual bases of adolescent development* (3rd ed., pp. 524–558). Hoboken, NJ: Wiley.
- Lerner, R. M., Bowers, E. P., Geldhof, J. G., Gestsdottir, S., & DeSouza, L. (2012). Promoting positive youth development in the face of contextual changes and challenges : The roles of individual strengths and ecological assets. *New Directions for Youth Development*, 135, 119–128. <https://doi.org/10.1002/yn.20034>
- Lerner, R. M. (2004). *Liberty: Thriving and civic engagement among American youth*. Thousand Oaks, CA: Sage Publications.
- Lopus, Jane S., Dwi Sulistyorini Amidjono, and Paul W. Grimes. 2019. "Improving Financial Literacy of the Poor and Vulnerable in Indonesia: An Empirical Analysis." *International Review of Economics Education* 32: UNSP-100168. [10.1016/j.iree.2019.100168](https://doi.org/10.1016/j.iree.2019.100168).
- Mathias, K., Singh, P., Butcher, N., Grills, N., Srinivasan, V., & Kermod, M. (2019). Promoting Social Inclusion for Young People Affected by Psychosocial Disability in India – a Realist Evaluation of a Pilot Intervention. *Global Public Health*, 14(12), 1718–1732. <https://doi.org/10.1080/17441692.2019.1616798>
- Meinck, et al. (2019). Interventions to reduce gender-based violence among young people living with or affected by HIV/AIDS in low-income and middle-income countries. *AIDS*, 33, 2219–2236.
- Nabunya, P., Phionah, N., Christopher, D., Apollo, K., Byansi, W., Miriam, M., et al. (2019). Assessing the Impact of an Asset-Based Intervention on Educational Outcomes of Orphaned Children and Adolescents: Findings from a Randomised Experiment in Uganda. *Asia Pac J Soc Work Dev*, 29(1), 59–69. <https://doi.org/10.1080/02185385.2019.1575271>

- O'Mara-Eves, A., Brunton, G., Oliver, S., et al. (2015). The effectiveness of community engagement in public health interventions for disadvantaged groups: A meta-analysis. *BMC Public Health*, 15, 129. <https://doi.org/10.1186/s12889-015-1352-y>
- Pittman, 1999. Student-generated analogies: Another way of knowing. *Journal Research Science Teaching* (36) 1 - 22 10.1002/(SICI)1098-2736(199901)36:1<1::AID-TEA2>3.0.CO;2-2.
- Powers, Margaret, M., Evangelides Alicia, & Offerdahl Kate (2014). Overcoming youth marginalization: conference report and policy recommendations. Global Policy Initiative.
- Rao, Neomi, and Lakshmi Lingam. "Smartphones, Youth and Moral Panics: Exploring Print and Online Media Narratives in India." *Mobile Media & Communication*, (2020): 205015792092226. 10.1177/2050157920922262.
- Roth & Brooks-Gunn. (2000). What do adolescents need for health development? Implications for youth policy. *Social Policy Report*, 14.
- Sikenyi, M. (2017). Does Kenya's Youth Enterprise Development Fund Serve Young People? *IDS Bulletin*, 48(3), 127–140. <https://doi.org/10.19088/1968-2017.131>
- Sosa-Rubi, S. G., Saavedra-Avendano, B., Claudia Piras, S., Van Buren, J., & Bautista-Arredondo, S. (2017). True Love: Effectiveness of a School-Based Program to Reduce Dating Violence Among Adolescents in Mexico City. *Prevention Science*, 18(7), 804–817. <https://doi.org/10.1007/s11121-016-0718-4>
- Ssewamala, F. M., Wang, J.-H., Neilands, T. B., Bermudez, L. G., Garfinkel, I., Waldfogel, J., et al. (2018). Cost-Effectiveness of a Savings-Led Economic Empowerment Intervention For. *Journal of Adolescent Health*, 62(1S), S29–S36. <https://doi.org/10.1016/j.jadohealth.2017.09.026>
- Stackpool-Moore, L., Bajpai, D., Caswell, G., Crone, T., Dewar, F., Gray, G., et al. (2016). Linking Sexual and Reproductive Health and Rights and HIV Services for Young People: The Link Up Project. *Journal of Adolescent Health*, 60(2), S3–S6. <https://doi.org/10.1016/j.jadohealth.2016.11.008>
- Stark, L., Seff, I., Assezenew, A., Eoomkham, J., Falb, K., & Ssewamala, F. M. (2018). Effects of a Social Empowerment Intervention on Economic Vulnerability for Adolescent Refugee Girls in Ethiopia. *J Adolesc Health*, 62(1S), S15–S20. <https://doi.org/10.1016/j.jadohealth.2017.06.014>
- Tirrell, J. M., John Geldhof, G., King, P. E., Dowling, E. M., Sim, A. T. R., Williams, K., et al. (2019). Measuring Spirituality, Hope, and Thriving Among Salvadoran Youth: Initial Findings from the Compassion International Study of Positive Youth Development. *Child & Youth Care Forum*, 48(2), 241–268. <https://doi.org/10.1007/s10566-018-9454-1>
- Tozan, Y., Sun, S., Capasso, A., Wang, J.-H., Neilands, T. B., Bahar, O. S., et al. (2019). Evaluation of a Savings-Led Family-Based Economic Empowerment Intervention for AIDS-Affected Adolescents in Uganda: A Four-Year Follow-up on Efficacy and Cost-Effectiveness. *PLoS ONE*, 14(12). <https://doi.org/10.1371/journal.pone.0226809>
- Esbroeck, V., Dirk, S. C., & Verhofstadt, E. (2016). *Evaluation of the My Rights My Voice Programme: Synthesis Report*. Kessel-Lo, Belgium: South Research.
- Vojt, G., Skivington, K., Sweeting, H., Campbell, M., Fenton, C., & Thomson, H. (2018). Lack of evidence on mental health and well-being impacts of individual-level interventions for vulnerable adolescents: Systematic mapping review. *Public Health*, 161, 29–32.
- Vyas, A. N., Nagaraj, N., Genovese, J., Malhotra, G., Dubey, N., Hingorani, R., et al. (2020). The Girl Rising: 'We Dream, We Rise' Social Media Campaign in India: Reach, Engagement and Impact. *Journal of Creative Communications*, 15(1), 106–124. <https://doi.org/10.1177/0973258619878354>
- Zhou, Z., Shek, D. T., & Zhu, X. (2020). The importance of positive youth development attributes to life satisfaction and hopelessness in mainland Chinese adolescents. *Frontiers in Psychology*, 11, 2599.
- Zipp, S. (2017). Sport for Development with 'At Risk' Girls in St. Lucia. *Sport in Society*, 20(12), 1917–1931. <https://doi.org/10.1080/17430437.2017.1232443>
- Dr. Gina Chowa** is a the Johnson-Howard-Adair Distinguished Professor at the University of North Carolina at Chapel Hill School of Social Work and founding director of GSDI. Dr. Chowa's research is in youth economic security, financial inclusion, and workforce development.
- Dr Rainier Masa** is an Assistant Professor at the University of North Carolina at Chapel Hill School of Social Work and lead researcher at GSDI. Dr. Masa's research is in youth health and economic security.
- Dr Neil Bilotta** is a Clinical Assistant Professor at the University of North Carolina at Chapel Hill School of Social Work and a researcher at GSDI. Dr Bilotta's research is in social inclusion, culturally responsive ethics of global social work, and youth participation.
- Ms Miranda Manzanera** is the Research Project Coordinator at GSDI at the University of North Carolina at Chapel Hill School of Social Work. Ms Manzanera is passionate about youth development and youth participation in their development.